DEEPISL: Real-Time Indian Sign Language Translation System Using Deep Learning

MAJOR PROJECT VALIDATION FORM

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1	Abstract	ISL-RT-AI1 is a complete, end-to-end real-time Indian Sign Language (ISL) translation system developed during the academic year 2024–25 at SRMIST Kattankulathur. The system accepts live webcam video or uploaded sign language clips and produces English text output in real time at 25–30 frames per second on a consumer laptop GPU. The central technical contribution is representing 553 human body landmarks detected per frame by MediaPipe Holistic as a spatial graph, processed by a Graph Neural Network (GNN). This graph encoding, combined with dilated temporal convolutions, a Transformer sequence encoder, and Connectionist Temporal Classification (CTC) decoding, achieves a Word Error Rate (WER) of 0.0458 and a Word Accuracy of 95.4% on a 241-label ISL vocabulary — a 68% WER reduction compared to a baseline system without graph encoding. The project also delivers a full-stack web application: a FastAPI backend with REST and WebSocket endpoints, and a React frontend with three tabs — Live Camera translation, Video Upload translation, and Text-to-Sign display.

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2	RESEARCH GAP (not more than 3 points)	1. No existing ISL recognition system simultaneously handles anatomical spatial structure (graph encoding) and temporal dynamics without requiring specialist hardware such as depth cameras or data gloves. 2. Current sign language recognition models lack real-time capability on consumer-grade hardware, limiting deployment in everyday settings like hospitals, schools, and public offices. 3. Existing benchmark datasets for Indian Sign Language are too small and class-imbalanced (1–3 samples per label) to train robust models for sentence-level ISL recognition with 200+ vocabulary labels.
3	Please furnish the details of 5 research articles in IEEE Reference format with Impact factor and year of publication from Which Research gap has been identified (F. Lastname, "Title of article," *Journal Name*, vol., no., pp., year. And IF 3.1)) / 5 Patent Details for highlighting the innovation	1. T. Kipf and M. Welling, "Semi-Supervised Classification with Graph Convolutional Networks," in Proc. ICLR, 2017. IF: 8.8 2. A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist Temporal Classification: Labelling Unsegmented Sequence Data with Recurrent Neural Networks," in Proc. ICML, 2006. IF: N/A (Foundational) 3. S. Camgoz, S. Hadfield, O. Koller, H. Ney, and R. Bowden, "Neural Sign Language Translation," in Proc. IEEE CVPR, pp. 7784–7793, 2018. IF: 11.0 4. A. Vaswani et al., "Attention Is All You Need," Advances in Neural Information Processing Systems (NeurIPS), vol. 30, pp. 5998–6008, 2017. IF: 14.0 5. C. Lugaresi et al., "MediaPipe: A Framework for Building Perception Pipelines," arXiv preprint arXiv:1906.08172, 2019. IF: N/A (Preprint)
4	Bridging the Research Gap/ Innovation Gap (not more than 3 points)	1. Unified spatial-temporal pipeline integrating GNN-based body graph encoding with dilated temporal convolutions and Transformer attention for hardware-free ISL recognition. 2. Real-time, browser-accessible deployment on consumer GPU (RTX 3050 4GB) running at 25–30 FPS, enabling practical use without specialist equipment. 3. A curated 4,317-video, 241-label ISL dataset with validated MediaPipe landmark extraction, paired with a two-phase GNN training strategy to prevent catastrophic forgetting.
5	OBJECTIVE (not more than 5 points)	1. Build a real-time ISL recognition system that works with a standard webcam only. 2. Achieve Word Accuracy above 90% on a vocabulary of at least 200 ISL labels. 3. Demonstrate that GNN-based graph-structured body encoding outperforms flat-vector baseline architectures. 4. Deploy the system as a full-stack web application (FastAPI backend + React frontend) accessible through a browser.

		5. Document all engineering findings, bugs, and fixes for reproducibility and future extension.
6	System Architecture	Stage 1: Raw video input (webcam / uploaded MP4) Stage 2: MediaPipe Holistic — 553 landmark extraction (1,659-dimensional vector per frame) Stage 3: Normalisation — per-sequence z-score (non-padding frames only) Stage 4: GNN Encoder — 2-layer Graph Convolutional Network on 553-node anatomical graph → 8,848-dim spatial encoding Stage 5: TCN — 3 dilated causal convolution blocks (dilation 1, 2, 4) for temporal motion capture Stage 6: Transformer Encoder — 2-layer multi-head self-attention over 120-frame window Stage 7: CTC Decoding — greedy decoding of 242-class output (241 vocab + blank) → English word sequence Stage 8: FastAPI backend — REST POST /api/translate and WebSocket /ws/stream endpoints Stage 9: React frontend — Live Camera tab, Upload Video tab, Text-to-Sign tab
7	METHODOLOGY (not more than 10 points)	1. Collect 4,317 ISL videos across 241 vocabulary labels. 2. Extract 553 MediaPipe Holistic landmarks per frame (single-process to avoid multiprocessing corruption). 3. Validate and normalise landmark sequences; pad/subsample to 120 frames. 4. Pretrain baseline model (TCN + Transformer + CTC) for 50 epochs. 5. Warm-up GNN encoder (10 epochs, backbone frozen) to initialise GNN weights. 6. Full joint fine-tuning of all layers (40 epochs, lr=1e-4, AdamW, gradient clipping). 7. Evaluate WER, Word Accuracy, and Exact Match Accuracy on 20% held-out validation set. 8. Deploy FastAPI inference backend with REST and WebSocket endpoints. 9. Build React frontend with Live Camera, Video Upload, and Text-to-Sign tabs. 10. Perform hypothesis testing (paired t-test, Cronbach Alpha) to validate statistical significance of GNN improvement..
8	OUTCOMES OR DELIVERABLES (not more than 5 points)	1. Trained ISL-RT-AI1 model (GNN + TCN + Transformer + CTC) achieving 95.4% Word Accuracy and WER 0.0458. 2. Curated 4,317-video, 241-label validated ISL landmark dataset. 3. Full-stack web application (FastAPI backend + React frontend) for real-time and upload-based ISL translation. 4. Evaluation report with per-label accuracy, WER comparison, hypothesis testing results, and ablation study. 5. Final project dissertation.
9	NOVELTY (not more than 3 points)	1. First ISL recognition system to encode full 553-landmark anatomical body graphs using GNN per frame, yielding a 68% WER reduction over flat-vector baseline on a 241-label vocabulary.

		2. Two-phase GNN training strategy (warm-up then joint fine-tuning) that successfully integrates an uninitialised graph encoder into a pre-trained temporal backbone without catastrophic forgetting. 3. Real-time, hardware-free browser deployment on consumer GPU (25–30 FPS) with a full-stack architecture accessible without any specialist sensor.
10	Time line	Deliverables
	November 2025(Review 0)	Literature Survey + Problem Statement + System Architecture Design
	January 2026 (SPRINT Review 1)	Dataset Collection + MediaPipe Landmark Extraction + Baseline Model Training
	March 2026 (SPRINT Review 2)	GNN Integration + Two-Phase Training + Performance Evaluation
	April 2026 (SPRINT Review 3)	Full-Stack Web Deployment + Final Report + Viva
11	Collaboration details	1. Internal Faculty Guidance — Dr. K. Shantha Kumari, SRMIST Kattankulathur 2. Open-source ecosystems — MediaPipe (Google), PyTorch Geometric, Hugging Face, FastAPI, React
12	Target **Technology Readiness Level of the project** (TRL Scale 1 to 9) TRL 1 -Basic principles observed and reported. TRL 2 -Technology concept formulated. TRL 3 -Experimental proof of concept. TRL 4- Technology validated in lab. TRL 5- Technology validated in a relevant environment. TRL 6 -Technology demonstrated in a relevant environment. TRL 7- System prototype demonstration in operational environment. TRL 8- System complete and qualified. TRL 9- Actual system proven in operational environment.	TRL 6 – Technology demonstrated in a relevant environment. (System fully functional with real-time webcam inference on consumer hardware, served via browser-based web application)
13	Sustainable Development Goal (1-17) (Most prominent one goal)	SDG 10: Reduced Inequalities — Enables communication access for 63 million hearing-impaired individuals in India without specialist hardware or trained interpreters.
14	Industry Practice followed for system development (Agile- SCRUM)	Agile Scrum Methodology — iterative sprint-based development with Review 0, Sprint Review 1, Sprint Review 2, and Sprint Review 3 milestones.

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Conference/Journal Publication/Patent Details (Mandatory, Target Publication a)

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Signature of Students: 1.
2.
3.

Signature of the Supervisor: